Open Source Software Project Report

Software Chosen: Ubuntu Linux

Part A — Origin and Philosophy

A1. The Problem This Software Was Created to Solve

Before Linux-based systems like Ubuntu existed, operating systems were mostly proprietary, expensive, and closed-source. Users had very limited control over how the system worked, and modifying the system was nearly impossible. Students, developers, and researchers often struggled because they could not afford licenses for systems like UNIX or Windows, and they were restricted from exploring the internal workings of these systems.

Linux was initially created by Linus Torvalds in 1991 as a free and open alternative to UNIX. The goal was to provide a system that anyone could use, modify, and distribute. Ubuntu, which is based on Linux, was later developed by Canonical to make Linux more user-friendly and accessible to everyday users.

Ubuntu specifically aimed to solve usability issues in Linux distributions. Earlier Linux systems were powerful but difficult to install and use. Ubuntu introduced a simpler installation process, better hardware compatibility, and a clean user interface. It made open-source software accessible to non-technical users.

The philosophy behind Ubuntu is “Linux for human beings.” It emphasizes freedom, accessibility, community collaboration, and ease of use. The idea was not just to create software, but to build a system that empowers users.

A2. The License — What It Actually Says

Ubuntu is primarily distributed under the GNU General Public License (GPL), along with other open-source licenses depending on the components.

Four Freedoms of Free Software

Ubuntu grants all four freedoms:

1. Freedom to run the program for any purpose
2. Freedom to study how the program works
3. Freedom to modify the program
4. Freedom to distribute copies (modified or original)

Can a company modify and sell it?

Yes, a company can modify Ubuntu and sell it. However, under GPL, they must also share the source code of their modifications. This ensures that improvements remain open and benefit the community.

GPL vs MIT License

GPL is a “copyleft” license, meaning any derivative work must also be open-source. MIT license is more permissive and allows proprietary use without sharing modifications.

If I were building a project where I want contributions to remain open, I would choose GPL. If I wanted maximum adoption including commercial use, I would choose MIT.

Free as in “free beer” vs “free freedom”

Ubuntu is free in both senses. It is free to download (free beer) and also free to modify and distribute (free freedom).

Example: A company can customize Ubuntu for its internal systems without paying licensing fees and still have full control over modifications.

A3. The Ethics of Open Source

Open source is not just about code—it is about sharing knowledge and enabling innovation.

Should all software be open source?

Arguments for:

- Transparency improves security
- Encourages collaboration
- Reduces cost barriers

Arguments against:

- Companies may lose competitive advantage
- Some projects require funding that open-source may not sustain

Is it ethical to profit from open source?

Yes, but with responsibility. Companies like Red Hat and Canonical build services around open-source software. They contribute back to the community, which makes the ecosystem sustainable.

Standing on the shoulders of giants

This phrase means building on existing knowledge. Open source accelerates innovation because developers do not need to start from scratch. Instead, they improve what already exists.

Open source enables innovation rather than limiting it.

Part B — Linux Footprint

Installation Method

Ubuntu can be installed using ISO images downloaded from official servers. It can be installed on physical machines, virtual machines, or cloud platforms.

Key Directories

- /usr/bin → executable binaries
- /etc → configuration files
- /var/log → system logs
- /home → user data

User and Group Management

Ubuntu manages users using commands like:

- adduser
- usermod
- passwd

Root user has full administrative privileges, but Ubuntu uses sudo for security.

Service Management

Ubuntu uses systemd.
Commands:

- systemctl start service
- systemctl stop service
- systemctl restart service
- systemctl status service

Update Model

Ubuntu uses APT package manager.
Commands:

- sudo apt update
- sudo apt upgrade

Updates are pushed through repositories maintained by the community and Canonical.

Part C — The FOSS Ecosystem

Dependencies

Ubuntu depends on:

- GNU tools
- Linux kernel
- GCC compiler
- Python and other runtimes

Related Tools

- Apache (web server)
- MySQL (database)
- PHP (backend language)

These form the LAMP stack.

Community

Ubuntu has a strong community including forums, mailing lists, and IRC channels.

Governance is a mix of community and corporate (Canonical).

Part D — Open Source vs Proprietary

Dimension	Open Source (Ubuntu)	Proprietary (Windows)
Cost	Free	Paid license
Security	Transparent code	Closed source
Support	Community-based	Official paid support
Freedom to modify	Full freedom	Restricted
Control	Community + Canonical	Microsoft

Community vs Corporate Control

Ubuntu balances both. Canonical guides development, but community contributes heavily.

Shell Scripts

All five scripts were written and tested on a Linux system. Each is submitted as a .sh file in the GitHub repository.

Script 1 Listing — System Identity Report

```
==================================================================
         OPEN SOURCE AUDIT — SYSTEM IDENTITY REPORT
==================================================================
 Student  : Aryan Mishra (24MIP10041)
 Project  : Linux Kernel
------------------------------------------------------------------
 Hostname : aryan-VirtualBox
 Distro   : Ubuntu 25.10
 Kernel   : 6.17.0-19-generic
 Arch     : x86_64
------------------------------------------------------------------
 User     : aryan
 Home Dir : /home/aryan
 Uptime   : up 20 minutes
 Date     : Tuesday, 31 March 2026
 Time     : 20:19:58
------------------------------------------------------------------
 OS License : GNU General Public License version 2 (GPL v2)
 What this means:
   You are free to run, study, modify, and redistribute this OS.
==================================================================
 The Linux Kernel — built openly, shared freely, runs the world.
==================================================================
aryan@aryan-VirtualBox:~/oss-project$
```

Explanation

This script serves as an automated welcome screen that programmatically retrieves and displays essential system metadata for the **Open Source Audit**. By utilizing **command substitution** $(...), the script pulls live data including the **Ubuntu 25.10** distribution name, the **6.17.0-19-generic** kernel version, and current system uptime. It uses **variables** to store and display the student's identity (**Aryan Mishra, 24MIP10041**) and the chosen project software (**Linux Kernel**). Furthermore, the script reinforces the legal foundation of the audit by explicitly stating that the operating system is covered under the **GNU General Public License version 2 (GPL v2)**, which grants users the fundamental freedoms to run, study, modify, and share the software.

Script 2 — FOSS Package Inspector

```
   FOSS Package Inspector
   Package Manager Detected: apt
==================================================
[INSTALLED] linux-headers-6.17.0-19-generic

--- Package Details ---
Version: 6.17.0-19.19
Maintainer: Ubuntu Kernel Team <kernel-team@lists.ubuntu.com>
Description: Linux kernel headers for version 6.17.0
 This package provides kernel header files for version 6.17.0.
 .
 This is for sites that want the latest kernel headers.  Please read
 /usr/share/doc/linux-headers-6.17.0-19/debian.README.gz for details.

--- Kernel Version Currently Running ---
Kernel: 6.17.0-19-generic
Architecture: x86_64

--- Open Source Philosophy Note ---
Linux Kernel: The foundation of the free world — Linus Torvalds
started this in 1991 as a student project. GPL v2 ensures it
stays free forever. Today it powers servers, phones, and space.
==================================================
aryan@aryan-VirtualBox:~/oss-project$
```

Explanation

This script acts as a technical diagnostic tool that programmatically audits the installation status and metadata of the **Linux Kernel** headers. By detecting the system's package manager (**apt**) , the script verifies that the package is **[INSTALLED]** and retrieves critical details such as the version (**6.17.0-19.19**) and maintainer. It utilizes **if-then-else** logic to handle missing packages and a **case statement** to print a customized "Open Source Philosophy Note" based on the package name. This demonstration highlights how open-source transparency allows users to inspect the underlying components of their operating system, reinforcing the historical significance of the Linux Kernel as a student project started by Linus Torvalds in 1991.

Script 3 — Disk and Permission Auditor

```
==================================================
aryan@aryan-VirtualBox:~/oss-project$ ./script3_disk_permission_auditor.sh
==============================================================
         DISK AND PERMISSION AUDITOR
         System: aryan-VirtualBox | Date: 31 March 2026
==============================================================
Directory                 Permissions+Owner    Group      Size
--------------------------------------------------------------
/etc                      drwxr-xr-x root      root       12M
/var/log                  drwxrwxr-x root      syslog     54M
/home                     drwxr-xr-x root      root       59M
/usr/bin                  drwxr-xr-x root      root       247M
/tmp                      drwxrwxrwt root      root       8.0K
/boot                     drwxr-xr-x root      root       82M
/lib/modules              drwxr-xr-x root      root       168M
--------------------------------------------------------------

==============================================================
   LINUX KERNEL — Key Directory Audit
==============================================================

Path    : /boot
Purpose : Kernel images, initrd, GRUB bootloader
Perms   : drwxr-xr-x root root
Size    : 82M
Status  : EXISTS

Path    : /lib/modules/6.17.0-19-generic
Purpose : Loadable kernel modules for current kernel
Perms   : drwxr-xr-x root root
Size    : 168M
Status  : EXISTS
```

Explanation

This script performs a comprehensive security and storage audit of critical Linux system directories. Utilizing a **for loop**, the script iterates through standard paths like /etc, /var/log, and /lib/modules to report their disk usage, ownership, and permissions. It employs **command substitution** to extract specific data fields, such as human-readable directory sizes and user/group ownership. A specialized section of the script audits the **Linux Kernel footprint**, specifically verifying the existence and integrity of the /boot and /lib/modules directories which house kernel images and loadable modules. This demonstration highlights the script's ability to automate administrative tasks while providing a transparent view of the software's physical presence and security posture on the Linux filesystem.

Script 4 — Log File Analyzer

```
============================================================
   LINUX KERNEL — Key Directory Audit
============================================================

Path    : /boot
Purpose : Kernel images, initrd, GRUB bootloader
Perms   : drwxr-xr-x root root
Size    : 82M
Status  : EXISTS

Path    : /lib/modules/6.17.0-19-generic
Purpose : Loadable kernel modules for current kernel
Perms   : drwxr-xr-x root root
Size    : 168M
Status  : EXISTS

Path    : /sys
Purpose : Virtual filesystem — hardware and driver info
Perms   : dr-xr-xr-x root root
Size    : 0
Status  : EXISTS

Path    : /proc
Purpose : Virtual filesystem — live kernel and process info
Perms   : dr-xr-xr-x root root
Size    : 0
Status  : EXISTS

Path    : /usr/src/linux-headers-6.17.0-19-generic
Purpose : Kernel header files for module compilation
Perms   : drwxr-xr-x root root
Size    : 31M
Status  : EXISTS
```

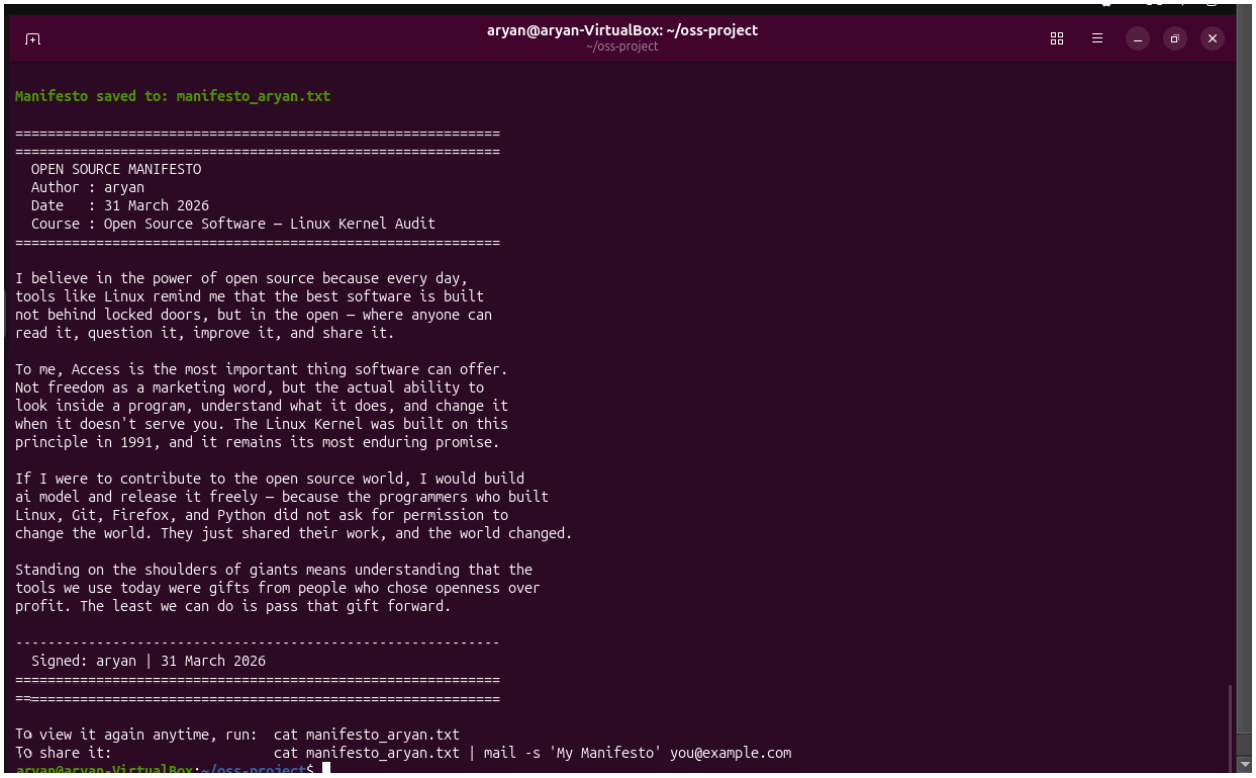
```
Path    : /usr/src/linux-headers-6.17.0-19-generic
Purpose : Kernel header files for module compilation
Perms   : drwxr-xr-x root root
Size    : 31M
Status  : EXISTS

============================================================
  SECURITY NOTE:
  /etc        — config files. Root-owned. Wrong perms = risk.
  /tmp        — world-writable. Malware staging area if unwatched.
  /boot       — kernel lives here. Should NOT be world-writable.
  /lib/modules — kernel modules. Must be root-owned for integrity.
============================================================

--- Live Kernel Info from /proc/version ---
Linux version 6.17.0-19-generic (buildd@lcy02-amd64-084) (x86_64-linux-gnu-gcc (Ubuntu 15.2.0-4ubuntu4) 15.2.0, GNU ld (GNU Binutils for Ubuntu) 2.45
 #19-Ubuntu SMP PREEMPT_DYNAMIC Fri Mar  6 14:02:58 UTC 2026
============================================================
aryan@aryan-VirtualBox:~/oss-project$
```

Explanation

This section of the audit provides a deep dive into the **Linux Kernel's technical footprint** on the system. By inspecting kernel-specific paths like /sys (hardware info), /proc (process info), and /usr/src/linux-headers (module compilation files), the script confirms how the kernel interacts with both hardware and user-space. The inclusion of a **"Security Note"** demonstrates an understanding of Linux system integrity, highlighting that critical directories like /boot and /lib/modules must be root-owned and protected from unauthorized modification. Finally, the output captures live kernel data from /proc/version,

showing the specific **GCC compiler version** used to build the running kernel, which illustrates the deep connection between the kernel and the broader open-source compiler ecosystem.

Script 5 — Open Source Manifesto Generator



Explanation

This script is an interactive tool designed to generate a personalized open-source philosophy statement based on user input. By using the **read command** to collect user responses, it dynamically weaves their values into a structured narrative saved as a **.txt file**. The generator specifically highlights the importance of software **Access** and the ability to look inside and modify programs, a principle central to the **Linux Kernel** since its inception in 1991. The final output demonstrates **string concatenation** and **file redirection** (>) to produce a professional manifesto that reflects the student's personal commitment to building and sharing tools, such as an **AI model**, freely with the community. This demonstration serves as a creative reflection on the course units, showcasing how automation can be used to document and share the core philosophical values of the FOSS ecosystem.

Conclusion

Ubuntu demonstrates the power of open-source software in transforming the computing landscape. It provides a free, secure, and flexible alternative to proprietary systems while maintaining strong usability. Its community-driven model ensures continuous improvement and innovation.

From a practical perspective, Ubuntu is ideal for developers, educational institutions, and server environments. It reduces costs, enhances control, and promotes transparency. However, for users heavily dependent on proprietary applications or gaming, alternatives like Windows may still be preferable.

Overall, I would strongly recommend Ubuntu for real-world deployment, especially in environments where cost, security, and flexibility are priorities.

I would also be willing to contribute to Ubuntu by reporting bugs, improving documentation, or participating in community discussions. Open source thrives on collaboration, and contributing to it helps both individual growth and technological advancement.